

Collecting a corpus of newly-published books. The only existing public dataset for book-length summarization is BookSum (Kryscinski et al., 2022), which contains famous books from the Project Gutenberg public-domain repository along with reference summaries scraped from popular websites such as CliffNotes and GradeSaver. Both the source books and reference summaries are in the pretraining data of existing LLMs: Chang et al. (2023) confirm that many books in the BookSum held-out split (e.g., *The Adventures of Huckleberry Finn*, *The Picture of Dorian Gray*) are among the most-memorized books by GPT-4 and GPT-3.5-Turbo, and we were able to auto-complete several reference BookSum summaries by prompting GPT-4 with a short prefix of the summary.

To reduce the confounding impact of summary memorization, we manually collect 100 books³ published within the past year to form our dataset (see Table 3 for a full list). Some of these books could still have appeared in the pretraining dataset of recent LLMs such as Claude 2 and LLaMa2, although it is much less likely than in BookSum. However, *summaries* of these books do not publicly exist: we did not find *summaries* online for any books in our dataset, which significantly lowers the possibility of LLM memorization.⁴ The average length of the books in our dataset is 190K tokens, compared to 112K tokens in BookSum. Due to copyright laws, we cannot publicly release this dataset; even if we could, we would still recommend that researchers collect their own datasets of newly-published books to minimize contamination with LLMs of the future.

An evaluation framework for book-length summarization. Since we lack gold summaries, we design our evaluation framework to be reference-free, which aids in scalability. To do this, our evaluation framework synthesizes best-practices of prior document understanding and summarization evaluation research. Our evaluation employs: (1) *fine-grained evaluation units* as recommended by LongEval (Krishna et al., 2023); (2) information-seeking questions to represent naturally-occurring *points of confusion* (Ko et al., 2020; Wu et al., 2023; Meng et al., 2023; Newman et al., 2023); and (3) focus on *summary coherence*, which evaluates the logical structure and readability of the summary itself (Goyal et al., 2022a). We do not directly evaluate the *faithfulness* of the summaries (i.e., how factually accurate they are at conveying information from the source text), as the length of the source texts poses considerable issues for any existing faithfulness evaluation. We qualitatively discuss faithfulness in Section 5 and leave further investigation for future work.

Annotation protocol. We implement our framework through a source- and reference-free annotation protocol where (1) annotators read through an LLM-generated summary, (2) highlight all confusing spans, and (3) ask question(s) for each marked span that highlight their confusion.⁵ See Table 1 (third column) for examples of spans and questions produced by our annotators. We hired four annotators with extensive English proofreading experience on Upwork⁶, each of whom annotated 25 disjoint summaries. Each summary takes roughly 30 minutes to fully annotate with spans and questions, and we paid \$15 USD per summary for a total of \$3K to evaluate both prompting strategies. To generate the summaries, we set the base LLM to GPT-4 with a chunk size of 4096 and a maximum summary length $G_n = 1200$; other hyperparameters are detailed in Section 5. In total, the annotators mark **840** (incremental updating) and **353** (hierarchical merging) coherence errors for GPT-4-generated summaries; see Table 1 (right) for the split across error types.

Validating the annotations: Typical measures of agreement are difficult to obtain in our setup, as measuring recall would require ground truth annotations with all possible coherence errors in the summaries; additionally, Goyal et al. (2022a) and Dou et al. (2022) observed low recall among annotators when evaluating machine-generated text at a fine-grained level. This motivates us to instead measure the *precision* of a given error annotation (i.e., after reading the corresponding question, do you agree that the span is confusing?), as it is simpler and cheaper while still being an informative metric. Given a span from a summary marked as containing an error, along with questions highlighting the confusion, we ask annotators (1) whether they think the span is confusing; and (2) whether the corresponding questions highlight the central confusion. We use the same four annotators hired before for this task, but make them validate human and (and later GPT-4) annotations for 25 books that they did *not* annotate in the first task. Overall, we validated 1,659 annotations for a total cost of

³We roughly balance our dataset across the following genres: fiction, non-fiction, sci-fi, fantasy, historical, contemporary, and memoir. We also include both linear and non-linear (multi-perspective and time-shifting) narratives in the dataset, and we purchase electronic copies of each of the 100 books in the dataset.

⁴However, we did find book reviews, which intentionally do not reveal major plot points or other spoilers.

⁵We also enabled forming relations between two spans in case multiple spans contributed to the same issue.

⁶<http://upwork.com>

Table 1: Definition of all coherence error types, an example annotation for each, and their prevalence (%) in generated summaries, which is calculated as the number of error occurrences in all summaries normalized by the total number of sentences in all summaries.

Error Type	Definition	Example spans & questions	% errors per sentence inc / hier
Entity omission	An entity (e.g., person, object, place) is mentioned in the summary, but key context or details are missing or unclear.	<i>Span:</i> A mysterious man introduces Proctor to "Arrivalism." <i>Question:</i> Who is this mysterious man?	7.3 / 3.71
Event omission	An event is mentioned in the summary, but key details are missing or unclear.	<i>Span:</i> During a mission to find Caeli, Proctor is captured by watchmen while Thea escapes. <i>Question:</i> What happened to Caeli?	4.25 / 2.27
Causal omission	A reason or motivation is missing or under-explained.	<i>Span:</i> Proctor seeks answers from... Callista about the investigation. <i>Question:</i> Why would Callista know something about the investigation?	2.75 / 1.21
Discontinuity	An interruption in the flow of the narrative such as sudden jumps in time or perspective.	<i>Span:</i> In the new settlement, Thea adjusts to her life, working hard and finding solace in nature. <i>Question:</i> Why the shift to Thea's perspective?	2.23 / 1.56
Salience	Inclusion of details that do not contribute to the main plot.	<i>Span:</i> His father... flees, resulting in a chaotic chase on the pier. <i>Question:</i> What is the significance of this incident?	1.42 / 1.03
Language	Spelling or grammar issues; ambiguous wording.	<i>Span:</i> Despite her love for him, Deborah is heartbroken by his decision. <i>Question:</i> Why is the preposition "Despite" used here when she is, in fact, heartbroken because of her love for him?	0.82 / 0.71
Inconsistency	A discrepancy or contradiction within a story's plot, character development, or themes.	<i>Span:</i> In a farewell, Proctor marries his brother Malcolm to Cynthia and says goodbye to his loved ones. <i>Question:</i> If Cynthia is his mother and Malcolm is his brother, how can a mother and son marry?	0.97 / 1.03
Duplication	Redundant repetition of similar information.	<i>Span 1:</i> Proctor... deals with students and school issues, seeking help from Callista to fund a roof replacement. <i>Span 2:</i> Proctor's life continues as he... deals with school issues, such as funding for a roof replacement <i>Question:</i> Why does the same information appear twice?	2.12 / 1.18

\$418.90 (USD),⁷ and we discover that **79.7%** of annotated spans are validated as legitimate through this task. More details on our validation can be found in Appendix J.

Categorizing coherence errors: After collecting spans and questions from the annotators, we develop an error taxonomy consisting of the eight types detailed in Table 1, which covers the vast majority of annotations, and we manually code each annotation using this taxonomy. We intentionally went through this process without relying on the SNaC taxonomy (Goyal et al., 2022a) so as to not be overly influenced by their error annotation schema which was tailor-made for fine-tuned summarization models. While we find considerable overlap in the two error schemas, we also discover two new instances of prominent errors not present in SNaC: *causal omissions* and *salience issues*. Our taxonomy also places less emphasis on language errors (e.g. coreference issues from SNaC) since modern LLMs rarely make such mistakes (Goyal et al., 2022b). Table 1 shows that omission errors are the most common across both incremental and hierarchical prompting strategies, and also that hierarchical merging makes fewer errors of every type but inconsistencies.

4 BOOOOKSCORE: AN AUTOMATIC EVALUATION METRIC

Since human evaluation of summary coherence is not scalable due to the high financial and time cost, we develop an automatic metric — BOOOOKSCORE — that prompts an LLM to identify instances of the eight error types we identified in Section 3. We validate BOOOOKSCORE via a human evaluation of its precision (following the annotation task discussed in the previous section) and show that its precision matches that of human annotators (78.2% vs. 79.7%). We then use BOOOOKSCORE to evaluate many other book-length summarization configurations, saving \$15K USD in evaluation costs and 500 hours in annotator time. We emphasize that incorporating definitions and examples from our error taxonomy into the prompt is critical to achieve high precision with BOOOOKSCORE.⁸

⁷This cost includes validation of both human and BOOOOKSCORE annotations.

⁸In preliminary experiments without definitions and few-shot demonstrations, we qualitatively observe significantly reduced annotation precision.

4.1 IMPLEMENTING BOOOOKSCORE

Motivated by prior successful efforts to evaluate LLM-generated text via LLMs, such as AlpacaEval (Dubois et al., 2023), FActScore (Min et al., 2023), and G-Eval (Liu et al., 2023b), BOOOOCScore automatically measures the coherence of summaries generated by a book-length summarization system via few-shot prompting. BOOOOCScore is both source-free and reference-free (i.e., it does not require access to the input book or a reference summary), similar to the SNaC classifier built for fine-tuned summarizers by Goyal et al. (2022a).

Specification: Assume we have a summary S consisting of sentences $s_1, s_2, \dots, s_n$.⁹ We develop a few-shot error-identification prompt E that instructs the LLM to identify any instances of one of the eight specified error types in a given sentence s_i of the summary. Concretely, we iterate over each sentence s_i in the summary, feeding the prompt E , full summary S , and target sentence s_i at each step. There are two acceptable outputs at each step: either (1) no error is found and the LLM outputs `No confusion`, or (2) an error(s) is identified and the LLM is asked to generate a corresponding question and associated error type. We include two full summaries with 42 sentence-level annotations in the prompt as demonstrations.¹⁰ The `BOOOOKSCORE` of a single summary S (Figure 2) is then computed as:

$$\text{BOOOOKSCORE}(S) = \frac{1}{n} \sum_{s_i \in S} [\text{LLM}(E, S, s_i) == \text{No confusion}] \quad (1)$$

When computing BOOOOKSCORE, we consider each sentence as a singular unit of confusion, rather than each of the questions associated with that sentence. This is because both LLMs and human annotators occasionally ask multiple questions that essentially target the same issue within a given sentence.¹¹ Thus, our metric intuitively measures the proportion of sentences in the summary that contain no errors (i.e., higher is better). To obtain a system-level score, we compute the mean BOOOOKSCORE across all summaries generated by that system.

Validating BOOOOKSCORE: We validate BOOOOKSCORE annotations in the same way that we validate human annotations in Section 3: by hiring human annotators to judge whether they agree with an LLM-generated annotation (here, GPT-4). We observe that the precision of human annotations is **79.7%**, while the precision of BOOOOKSCORE annotations is **78.2%** (details in Appendix J). Additionally, we compute BOOOOKSCORE using *human annotations* instead of LLM-generated ones for both GPT-4 configurations (i.e., replacing LLM(E, S, s_i) in Equation 1 with the human error annotation for s_i) and observe extremely similar system-level scores. Using human annotations in Equation 1 yields a BOOOOKSCORE of **82.1** and **89.4**¹² for

⁹After iterating over the design in numerous preliminary experiments, we find that our prompt works most reliably at the sentence level, rather than at the full summary level. As such, sentence tokenization is a required preprocessing step for BOOOOKSCORE. Future work should focus on implementations at the summary level, as it would save many calls to the LLM: here, we need to prompt the model separately for each sentence.

¹⁰These examples contain a combination of sentences with and without confusion, all the while maintaining a diverse range of error types. The full prompt can be found in M.4.

¹¹For example, the questions “Who is John? Is he Lia’s husband?” both seek to establish John’s identity. Counting the number of questions instead of highlighted sentences would inadvertently overstate the weight of certain errors found within the same sentence.

¹²Recall that human annotators can (1) highlight multiples consecutive sentences as one span and (2) create relations between two spans, while GPT-4 can only highlight single sentences as spans. To adjust for this difference, we treat both consecutive sentences and relations as single sentences when computing BOOOOOScore for humans.

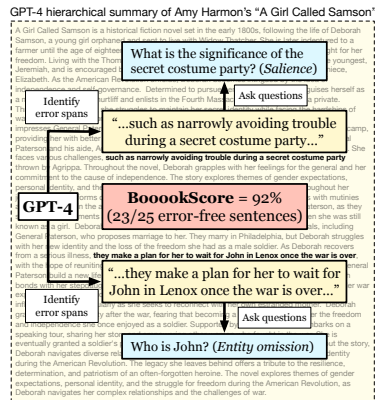


Figure 2: BOOOOKSCORE measures the proportion of error-free sentences in a summary, where coherence errors are detected by prompting GPT-4.